BIO 280 PATHOPHYSIOLOGY · JOYCE UNIVERSITY

Dr. Garg's Exam 2 Master Study Guide

Modules 3 & 4 — Cell Injury, Cancer, Inflammation, Pain, Wounds & Disease States

ⓘ What Exam 2 covers

This unit is about **how cells get hurt, how the body responds, and what happens when those responses go wrong**. **Module 3** follows the cell's story: injury → adaptation → cancer → inflammation → sepsis. **Module 4** applies it to real disease: pain, wound healing, neurodegenerative diseases, autoimmune/inflammatory conditions, and neuromuscular junction disorders. Exam format: **50 questions (multiple choice + select-all-that-apply)**, proctored with Respondus LockDown Browser.

ⓘ How this guide works

- **Click any topic card** in the navigator to jump to that section. Click a title bar to expand or close it.
- **All answers are hidden** — try first, then click "Show answer." Practice questions include both **multiple choice** and **select-all-that-apply (SATA)**, like the real exam.
- **Each topic has:** clear explanation, a mnemonic, a "common confusion" warning, practice questions, a clinical scenario, and a self-check.
- **Color code:** **purple = Cell Injury / Cancer / Inflammation**, **orange = Pain & Wounds**, **indigo = Neurodegenerative**, **teal = Autoimmune & NMJ diseases**.
- **To save as PDF:** use the button above, or press *Ctrl+P* / *Cmd+P* → "Save as PDF." All sections expand for printing.

□Suggested 7-Day Study Schedule ▼

Day	Focus
Day 1	Topics 1–2 (cellular injury, ischemia/hypoxia, the 5 adaptations, free radicals).
Day 2	Topics 3–4 (cancer: naming, benign vs malignant, grading, staging, metastasis, screening).
Day 3	Topics 5–6 (inflammation, histamine, neutrophils, opsonization, SIRS, sepsis).
Day 4	Topics 7–8 (pain types & pathway, wound healing phases & outcomes).
Day 5	Topics 9–10 (neurodegenerative: Parkinson's, Alzheimer's, Huntington's, MS, ALS, Guillain-Barré).
Day 6	Topics 11–12 (autoimmune/inflammatory diseases, neuromuscular junction diseases).
Day 7	Cheat sheet + disease comparison tables. Self-test with hidden answers.

□Topic Navigator

TOPIC 1 · MODULE 3
Cellular Injury, Ischemia & Hypoxia

TOPIC 2 · MODULE 3
Cellular Adaptations & Free Radicals

TOPIC 3 · MODULE 3
Cancer I: Carcinogens, Naming, Benign vs Malignant

TOPIC 4 · MODULE 3
Cancer II: Grading, Staging, Metastasis, Screening

TOPIC 5 · MODULE 3
Inflammation: Histamine, Neutrophils, Opsonization

TOPIC 6 · MODULE 3
SIRS, Sepsis & Systemic Inflammation

TOPIC 7 · MODULE 4

TOPIC 8 · MODULE 4

TOPIC 9 · MODULE 4

Pain: Types, Pathway & Subjectivity

Wound Healing: Phases & Outcomes

Neurodegenerative I: Parkinson's, Alzheimer's, Huntington's

TOPIC 10 · MODULE 4

Neurodegenerative II: MS, ALS, Guillain-Barré

TOPIC 11 · MODULE 4

Inflammatory & Autoimmune Diseases

TOPIC 12 · MODULE 4

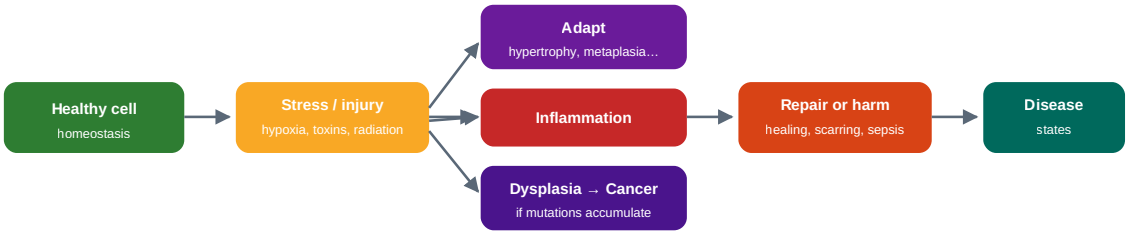
Neuromuscular Junction Diseases

□Exam Cheat Sheet

□Disease Comparison

□Glossary

□The Big Picture: The Cell's Journey



Module 3 = the cell's response · Module 4 = the diseases that result

TOPIC 1

□Cellular Injury, Ischemia & Hypoxia

MUST KNOW ▼

Cellular injury happens when severe stress prevents a cell from maintaining **homeostasis**. If the stress is mild or brief, the cell can recover; if it's severe or prolonged, the cell dies.

Causes of Cellular Injury

Many things can injure cells, including **lack of oxygen (hypoxia)**, **radiation**, **medications/toxins**, infections, physical trauma, extreme temperature, and nutritional imbalances. But one cause stands above the rest:

□Hypoxia is the #1 cause of cellular injury

Hypoxia vs. Hypoxemia vs. Ischemia vs. Anemia

Term	Means
Hypoxia	Low oxygen *in the tissues/cells* (not enough O&sub2; reaching cells).
Hypoxemia	Low oxygen *in the blood* specifically.
Ischemia	A *short-term lack of blood flow* (and therefore O&sub2;) to a tissue, **without cell death**. If it continues, it leads to infarction (tissue death).
Anemia	Too few RBCs / low hemoglobin to carry oxygen.

Common causes of hypoxia include **atherosclerosis** (narrowed vessels), **anemia**, and **ischemic stroke**.

How Hypoxia Hurts the Cell

Without oxygen, the cell can't run aerobic metabolism, so it switches to **anaerobic metabolism** — which makes far less ATP and produces lactic acid. The fallout:

- ↓ **ATP production** — the energy crisis behind everything else.
- **Sodium-potassium (Na^+/K^+) pump failure** — the pump needs ATP. When it fails, Na^+ and water rush in → **cellular swelling/edema**.
- **Inability to synthesize protein** — ribosomes detach; the cell can't build what it needs.

- If prolonged → irreversible damage and cell death.

Mnemonic: "**No O&sub2; = No ATP = No Pump = Puffy cell**." And for the look-alikes: hypox**EMIA** = in the bl**EMIA**…blood; hypox**IA** = in the t**IA**ssue. **Ischemia** = "I-stopped-the-stream" (blood flow stops, but cell still alive—for now).

⚠ **Common confusion:** **Ischemia** is reversible (short-term, no cell death yet); **infarction** is the tissue death that results if ischemia isn't relieved. Also, hypoxemia (blood) can cause hypoxia (tissue), but they aren't the same word.

Practice Questions

1. What is the #1 cause of cellular injury?

   Show answer

   Hypoxia (lack of oxygen).

2. Define ischemia.

   Show answer

   A short-term lack of blood flow/oxygen to a tissue, without cell death (reversible if corrected).

3. **SATA:** Which are potential impacts of hypoxia on a cell? (A) Na^+/K^+ pump failure (B) Increased cell size from more protein (C) Inability to synthesize protein (D) Decreased ATP production (E) Cellular swelling

   Show answer

   A, C, D, E. Hypoxia causes pump failure, loss of protein synthesis, decreased ATP, and swelling. It does NOT increase cell size from protein (the opposite — synthesis stops).

4. A patient has low oxygen levels measured in an arterial blood sample. Is this hypoxia or hypoxemia?

   Show answer

   Hypoxemia (low oxygen in the blood). It can lead to tissue hypoxia.

Clinical Scenario

A patient has a clot blocking a coronary artery. The heart muscle downstream

stops getting blood but hasn't died yet, and chest pain resolves when the clot is dissolved quickly. What was happening to that tissue, and what would have happened if the clot stayed?

Show answer

The tissue was **ischemic** — deprived of blood/O&sub2; but still alive (reversible). Dissolving the clot restored flow before cell death. If the clot had stayed, ongoing ischemia would progress to **infarction** (a heart attack — permanent muscle death).

☑ **Self-check:** Can you name the #1 cause of cell injury, distinguish hypoxia/hypoxemia/ischemia/anemia, and list 3 effects of hypoxia on the cell? If yes, move on.

TOPIC 2

Cellular Adaptations & Free Radicals

MUST KNOW

When cells face ongoing stress, they don't just give up — they **adapt** to survive. There are five classic adaptations. Learn the definition, an example, and whether it's reversible for each.

The 5 Cellular Adaptations

Adaptation	What changes	Example
Hypertrophy	Cells get **bigger** (size ↑), more functional but higher energy demand.	Left ventricle enlarges from hypertension; muscle growth from exercise.
Atrophy	Cells get **smaller** (size ↓), less energy demand.	Muscle wasting from disuse (cast), loss of nerve supply, poor nutrition.
Hyperplasia	Cells **increase in number** (only cells that can divide).	Breast gland cells during pregnancy; liver regeneration.
Metaplasia	One mature cell type is **replaced by another**. **Reversible**.	Chronic GERD: lower esophagus cells change to protect against acid (Barrett's).
Dysplasia	**Deranged/disordered** growth; abnormal size, shape, organization. **Pre-cancerous**.	Cervical dysplasia (abnormal PAP), esophageal dysplasia, dysplastic moles.

Causes can be physiologic (normal, like pregnancy or exercise) or pathologic (disease, like hypertension or chronic irritation). **Dysplasia** is the worrying one — it's disordered growth that can progress to cancer if the irritation continues.

Free Radicals

Free radicals are **unstable, highly reactive molecules** that cause **oxidative damage** to cells (they steal electrons from DNA, proteins, and membranes). They are produced by normal metabolism but increase with exposure to **carcinogens** (radiation, pollution, smoke). **Antioxidants** (like vitamins C and E) counteract them by neutralizing the unstable molecules.

Mnemonic: The adaptations — "**HHHMD**": **H**ypertrophy (bigger), **H**yperplasia (more), **H**ypotrophy/**A**trophy (smaller), **M**etaplasia (swap type), **D**ysplasia (disordered = danger). "**Trophy = size, Plasia = number/type.**" **Dysplasia = Dangerous (pre-cancer).**

⚠ **Common confusion:** **Hypertrophy** = bigger cells (size); **hyperplasia** = more cells (number). "-trophy" refers to size, "-plasia" refers to number or type. **Metaplasia** (reversible cell-type swap) is NOT the same as **dysplasia** (disordered, pre-cancerous growth).

Practice Questions

1. A weightlifter's muscle cells enlarge. Which adaptation is this?

   Show answer

   Hypertrophy (increase in cell *size*).

2. A patient with chronic acid reflux has esophageal cells that change into a different, more protective cell type. Name the adaptation and state whether it's reversible.

   Show answer

   Metaplasia — reversible if the acid reflux is controlled with treatment.

3. Which adaptation is considered pre-cancerous?

   Show answer

   Dysplasia (deranged, disordered cellular growth).

4. **SATA:** Which statements about free radicals are true? (A) They are stable molecules (B) They are unstable and highly reactive (C) They cause oxidative damage (D) They increase with carcinogen exposure (E) Antioxidants neutralize them

   Show answer

   B, C, D, E. Free radicals are *unstable* (not stable), so A is false.

Clinical Scenario

A long-time smoker's routine bronchoscopy shows the normal lining of the airway has been replaced by a tougher cell type, and a small area shows disordered,

abnormal cells. Name the two adaptations described, and explain which one is the bigger concern.

Show answer

The cell-type replacement is **metaplasia** (an adaptive, reversible response to chronic smoke irritation). The disordered abnormal area is **dysplasia** — the bigger concern, because it's pre-cancerous and can progress to carcinoma if smoking continues.

☑ **Self-check:** Can you define all 5 adaptations with an example each, explain "-trophy vs -plasia," and describe what free radicals do? If yes, move on.

TOPIC 3

Cancer I: Carcinogens, Naming & Benign vs Malignant

MUST KNOW

Cancer (neoplasia) is uncontrolled, abnormal cell growth. It begins when something damages a cell's DNA enough to remove the normal "brakes" on growth.

Carcinogens & Oncogenic Organisms

A **carcinogen** is any substance that promotes cancer by **damaging DNA** and creating mutations. Examples: **radiation** (UV sunlight, CT scans, X-rays), **tobacco smoke**, certain chemicals, and processed meats (e.g., bacon/nitrates). Some **infectious organisms are oncogenic** (cancer-causing):

- **HPV** (human papillomavirus) → cervical & other cancers (why the HPV vaccine matters).
- **Hepatitis B & C** → liver cancer.
- **HSV / Epstein-Barr virus** and *H. pylori* bacteria (stomach cancer) are also linked.

Naming Cancer by Tissue of Origin

Cancer name	Tissue of origin
Carcinoma	Epithelial cells (skin, gland & organ linings) — the most common type.
Sarcoma	Connective tissue (bone, muscle, fat, cartilage).
Glioma	Neural/glial tissue (brain & spinal cord).
Lymphoma	Lymphatic tissue.
Melanoma	Pigmented cells (melanocytes).
Myeloma	Plasma cells.
Leukemia	White blood cells (blood-forming tissue).

Benign vs. Malignant Tumors

Feature	Benign	Malignant (cancer)
Growth	Slow	Rapid, uncontrolled
Differentiation		

	Well differentiated (looks normal)	Poorly differentiated (looks abnormal)
Spread (metastasis)	Does NOT metastasize; stays local	CAN metastasize & invade nearby tissue
Capsule	Often encapsulated	Invasive, not encapsulated
Angiogenesis	Minimal	Induces its own blood vessels

The classic **hallmarks of cancer**: sustain their own growth signals, evade growth suppressors, resist cell death (apoptosis), induce angiogenesis (grow blood supply), avoid immune destruction, and enable limitless replication + metastasis.

> **Mnemonic:** "**Carcinoma = Cover** (epithelium covers/lines), **Sarcoma = Support** (connective tissue supports), **Glioma = Glia** (brain)." Malignant = "**M for Moves**" (metastasizes); benign = "**B for stays-Back**." Carcinogens "**Damage DNA**."

> ⚠ **Common confusion:** **Well differentiated = looks like normal tissue = less aggressive**. **Poorly differentiated = looks abnormal/immature = more aggressive (malignant)**. Students often flip these. The key word for malignancy is the ability to **metastasize**.

Practice Questions

1. A cancer of the brain/spinal cord glial tissue is called a:

   Show answer

   > Glioma.

2. Name two oncogenic viruses.

   Show answer

   > Any two of: HPV, Hepatitis B, Hepatitis C, HSV, Epstein-Barr virus.

3. **SATA:** Which are characteristics of a malignant tumor? (A) Rapid, uncontrolled growth (B) Well differentiated (C) Poor differentiation (D) Ability to metastasize (E) Can create its own blood vessels

Show answer

A, C, D, E. Malignant tumors are poorly differentiated, so B (well differentiated) is benign-like and incorrect.

4. What does a carcinogen do at the molecular level?

Show answer

Damages DNA, creating mutations that can lead to uncontrolled growth.

Clinical Scenario

A patient is told their renal cell carcinoma is malignant. They ask, "Does that mean it can spread?" How should the nurse explain the difference between their tumor and a benign one?

Show answer

Yes — "malignant" means the tumor grows rapidly, is poorly differentiated, can invade surrounding tissue, build its own blood vessels (angiogenesis), and **metastasize** to distant sites. A benign tumor grows slowly, stays local/ encapsulated, and does not spread.

☑ **Self-check:** Can you match cancer names to tissues (carcinoma/sarcoma/glioma…), list oncogenic viruses, and contrast benign vs malignant? If yes, move on.

TOPIC 4

Cancer II: Grading, Staging, Metastasis & Screening

MUST KNOW

Grading vs. Staging — the #1 thing to keep straight

	GRADING	STAGING
Looks at	How the **cells look** under the microscope (differentiation).	How far the cancer has **spread** in the body.
Key idea	Well differentiated (low grade) → poorly differentiated (high grade = worse).	Local → invasion → lymph nodes → distant metastasis.
System	Grade 1–4 by cell appearance.	**TNM** (Tumor size, Node involvement, Metastasis) and Stages I–IV.

The Stages (how far it has spread)

- **Stage I** — local; confined to the tissue of origin.
- **Stage II** — larger / local invasion into nearby tissue.
- **Stage III** — spread to **regional lymph nodes**.
- **Stage IV** — **distant metastasis** to other organs (most advanced).

Common Sites of Metastasis

Cancer commonly spreads to the **brain, bones, lungs, and liver**. (For many solid tumors, **brain and bones** are classic exam answers.) Spread happens through blood and lymphatic vessels.

Early Warning Signs of Cancer

General, early signs include **unexplained/unintentional weight loss**, **excessive fatigue/ weakness**, **random fevers**, **loss of appetite**, unusual bleeding (e.g., hematuria), a sore that won't heal, or a new lump. (Note: night sweats can occur but the classic early "general" set is weight loss, fatigue, fevers, appetite loss.)

Cancer Screening (what each test screens for)

Screening test	Screens for

Mammogram	Breast cancer
PAP smear	Cervical cancer (detects dysplasia)
Colonoscopy	Colorectal cancer
PSA (prostate-specific antigen)	Prostate cancer

Mnemonic: "**Grade = Glance** (how cells look); **Stage = Spread** (how far it traveled)." Stages climb with spread: I local → IV distant. **TNM = Tumor, Node, Metastasis.** Screening: "**P**AP = cervix, **M**ammogram = breast, **C**olonoscopy = colon, **PSA** = prostate."

⚠ **Common confusion: Grading looks at cells (microscope); staging looks at spread (body).** Poorly differentiated = a grading description, NOT a stage. There's no "Stage 5" — staging goes I–IV.

Practice Questions

1. A breast cancer has spread to the axillary (armpit) lymph nodes. What stage does this suggest?

   Show answer

   Stage III (regional lymph node involvement).

2. A lung cancer is described as "poorly differentiated." Is this grading or staging?

   Show answer

   Grading (it describes how the cells look under the microscope).

3. A PAP smear showing dysplasia is screening for which cancer?

   Show answer

   Cervical cancer.

4. **SATA:** Which are general early warning signs of cancer? (A) Unintentional weight loss (B) Excessive fatigue (C) Random fevers (D) Loss of appetite (E) Improved energy

   Show answer

   A, B, C, D. Improved energy is not a warning sign.

Clinical Scenario

A 50-year-old with a family history of breast cancer asks the nurse what screening she should get. What's the recommended test, and how does grading differ from the staging she's read about online?

Show answer

A **mammogram** screens for breast cancer. If cancer is found, **grading** tells how abnormal the cells look under the microscope (differentiation), while **staging** (TNM, Stages I–IV) tells how far it has spread — two different measures of severity.

☑ **Self-check:** Can you contrast grading vs staging, describe Stages I–IV, name common metastasis sites, and match screening tests to cancers? If yes, move on.

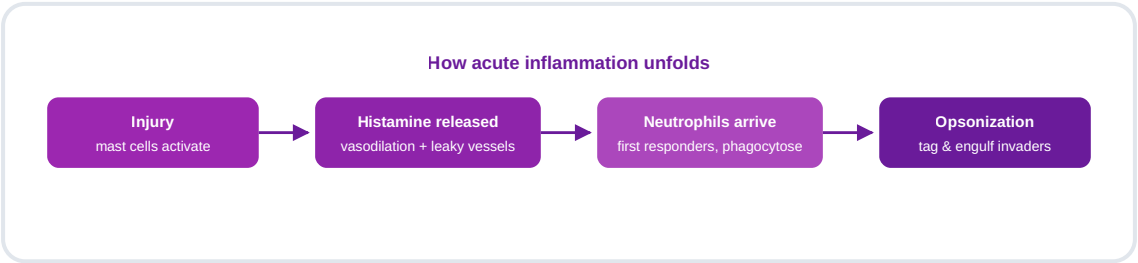
TOPIC 5

Inflammation: Histamine, Neutrophils & Opsonization

MUST KNOW

Inflammation is the body's protective response to injury or infection. Its purposes: **destroy the offending agent, wall off/localize the area, turn on the immune system, and stimulate tissue healing.**

The Classic Inflammation Sequence



Key Players

- **Mast cells** — the primary **histamine** releasers.
- **Histamine** — causes **vasodilation** and **increased capillary permeability**. Vasodilation brings more blood → **redness and heat**; leaky vessels let fluid out → **swelling**.
- **Neutrophils** — the **first cells** to arrive from the bloodstream; they phagocytose (engulf) invaders.
- **Macrophages** — engulf antigens that have been tagged by opsonization.
- **Cytokines** — chemical messengers between WBCs; **prostaglandins** stimulate fever & pain; **leukotrienes** speed the response.
- **Opsonization** — the process of **tagging antigens** so neutrophils and macrophages can find and engulf them more easily.

The Cardinal Signs (and why they occur)

Sign	Why
Redness (rubor)	Vasodilation (from histamine) brings more blood to the area.
Heat (calor)	Increased blood flow.
Swelling (tumor)	Increased permeability lets fluid leak into tissue.
Pain (dolor)	Pressure from swelling + chemical mediators (prostaglandins) on nerves.

Loss of function	Combined effect of swelling and pain.

Acute vs. Chronic Inflammation

	Acute	Chronic
Onset/ duration	Immediate; lasts days	Persistent; weeks to months/years
Main cells	Neutrophils (first), basophils, macrophages	Macrophages, lymphocytes
Cause	Injury, acute infection	Persistent infection, autoimmune disease (RA, lupus)
Outcomes	Resolution, abscess, or healing	Tissue damage, scarring & fibrosis

Outcomes of inflammation: resolution (back to normal), abscess/ulceration (ongoing), or scarring & fibrosis (tissue replaced by non-functional fibrous tissue — common in chronic/ autoimmune conditions).

> **Mnemonic:** The cardinal signs = "**Red Hot Swollen Painful**" (+ loss of function). **Neutrophils = first responders** ("the 9-1-1 cell" — **N**eutrophil = **N**ow). **Opsonization = "Op = Ohh, Please eat me"** (tagging for engulfment). Histamine → "**vasodilation + leaky**."

> ⚠ **Common confusion:** **Redness comes from vasodilation** (more blood), not from "direct nerve injury." And **neutrophils** (not lymphocytes or macrophages) are the FIRST to arrive in acute inflammation. Chronic inflammation shifts toward macrophages and lymphocytes.

Practice Questions

1. Why does redness occur at a site of acute inflammation?

   Show answer

   > Histamine causes vasodilation, increasing blood flow to the area.

2. What is opsonization?

Show answer

Tagging antigens so neutrophils and macrophages can more easily recognize and engulf them.

3. Which WBC is first to leave the bloodstream and arrive at the site of injury?

Show answer

Neutrophils.

4. **SATA:** Which are true of acute inflammation? (A) Turns on the immune system (B) Resolution, abscess, or healing are possible outcomes (C) Usually results from persistent disease (D) Neutrophils are first responders (E) It always becomes systemic

Show answer

A, B, D. Persistent disease describes chronic inflammation (C false), and acute inflammation does not always go systemic (E false).

Clinical Scenario

A patient sprains an ankle. Within minutes it's red, warm, swollen, and painful. Walk through which mediator and cells cause each sign.

Show answer

Mast cells release **histamine** → vasodilation (redness, heat) and increased permeability (swelling). Prostaglandins + pressure on nerves cause pain. **Neutrophils** arrive first to clean up damaged tissue, followed by macrophages. This is acute inflammation working as designed.

☑ **Self-check:** Can you explain histamine's effects, name the first responder cell, define opsonization, and contrast acute vs chronic inflammation? If yes, move on.

TOPIC 6

SIRS, Sepsis & Systemic Inflammation

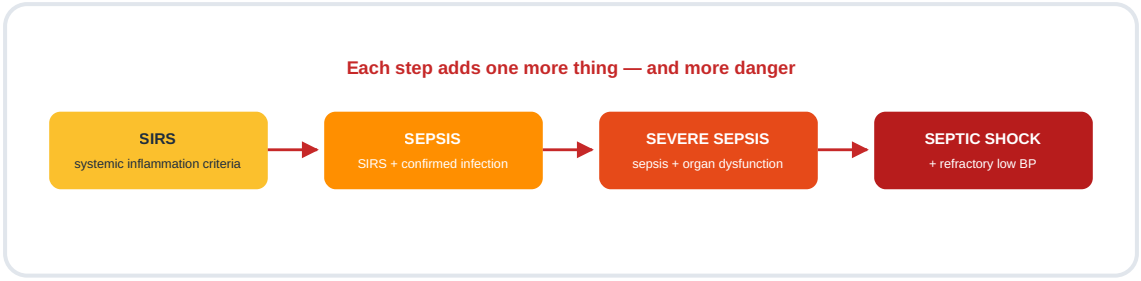
MUST KNOW

Sometimes inflammation isn't just local — it spreads **body-wide (systemic)**. This is driven by chemical mediators (cytokines, prostaglandins, leukotrienes) entering the bloodstream. The danger ladder runs **SIRS → Sepsis → Severe Sepsis → Septic Shock.**

Localized vs. Systemic Inflammation

	Localized	Systemic
Where	At the injury site only	Whole body
Signs	Redness, heat, swelling, pain, loss of function	Fever, ↑ WBC count, abnormal vital signs (↑HR, ↑RR), malaise

The Ladder: SIRS → Septic Shock



- **SIRS (Systemic Inflammatory Response Syndrome)** — the criteria used to **monitor the impact of inflammation on vital signs**. Includes abnormal temperature, heart rate, respiratory rate, and WBC count. SIRS can be from infection OR non-infection (burns, trauma, pancreatitis).
- **Sepsis** = SIRS **+ a confirmed/suspected infection**. Example: high HR, fever, high neutrophils, AND a chest x-ray showing pneumonia.
- **Severe sepsis** = sepsis **+ signs of end-organ damage** (e.g., altered mental status, low urine output, rising lactate).
- **Septic shock** = severe sepsis **+ dangerously low blood pressure** that doesn't respond to fluids (organ hypoperfusion).

Mnemonic: Build the ladder by **adding one word each step**: SIRS (just the vital-sign criteria) → **+infection** = Sepsis → **+organ damage** = Severe sepsis → **+low BP** = Septic shock. "**Sepsis = SIRS with a Source**."

⚠ **Common confusion:** The thing that turns **sepsis into septic shock** is **persistent low blood pressure / organ hypoperfusion** — not just a high WBC or fast breathing (those are already in SIRS/sepsis). And SIRS alone does NOT require an infection.

Practice Questions

1. What does SIRS stand for and what is it used for?

   Show answer

   > Systemic Inflammatory Response Syndrome — criteria to monitor the impact of inflammation on a client's vital signs.

2. A patient has fever, high HR, elevated neutrophils, and a chest x-ray showing pneumonia. What's the term?

   Show answer

   > Sepsis (SIRS criteria + a confirmed infection).

3. What additional finding distinguishes septic shock from sepsis?

   Show answer

   > Persistent low blood pressure / signs of end-organ hypoperfusion that don't respond to fluids.

4. **SATA:** Which signs would you expect in systemic inflammation? (A) Abnormal vital signs (B) High WBC count (C) Elevated body temperature (D) Redness limited to one fingertip (E) Fever

   Show answer

   > **A, B, C, E.** Redness at one fingertip is *localized*, not systemic.

Clinical Scenario

A patient with a urinary tract infection arrives with fever, HR 120, RR 24, high WBCs, and now confusion with a falling blood pressure that doesn't improve after IV fluids. Where on the SIRS→shock ladder are they?

Show answer

They started with sepsis (SIRS + infection). Confusion = organ dysfunction = **severe sepsis**; the fluid-refractory low BP pushes them into **septic shock** — a medical emergency.

☑ **Self-check:** Can you define SIRS, build the sepsis ladder (adding infection → organ damage → low BP), and list signs of systemic inflammation? If yes, you've finished Module 3!

TOPIC 7

Pain: Types, Pathway & Subjectivity

MUST KNOW

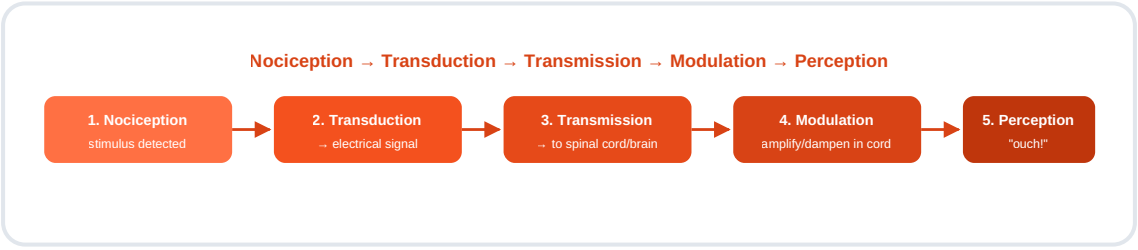
Acute vs. Chronic Pain

	Acute Pain	Chronic Pain
Purpose	**Protective** — warns of actual/ potential tissue damage	**Non-protective** — serves no useful warning
Timing	Immediate, with injury; self-limiting	Lasts > 3–6 months; ongoing/ recurrent
Correlates w/ injury?	Yes — correlates with severity of damage	No — does NOT correlate with original injury
Sequelae	Resolves as tissue heals	Depression, anxiety, fatigue, sleep issues

Nociceptive vs. Neuropathic Pain

Type	Source	Examples
Nociceptive — Somatic	Skin, soft tissue, muscle, bone	Femur fracture, ankle sprain (well-localized, aching)
Nociceptive — Visceral	Internal organs	Pancreatitis, spleen laceration (may have referred pain)
Neuropathic	Injury to PNS or CNS nerves	Diabetic neuropathy, Guillain-Barré, carpal tunnel, shingles (burning, tingling, shooting)

The Pain Pathway (5 steps, in order)



Why Is Pain Subjective?

Pain is whatever the patient says it is. Two people with the same injury can feel very different pain because of differences in **neurotransmitter levels (substance P, glutamate)**, prior experiences, emotions, culture, and the **modulation** step where the signal can be amplified or dampened in the spinal cord. This is why pain can't be measured objectively like blood pressure.

> **Mnemonic:** Pain pathway = "**NewtsTtransmit Their Mild Pain**" → **N**ociception, **T**ransduction, **T**ransmission, **M**odulation, **P**erception. **Acute = protective** ("acute = a cue"); chronic = lingers > 3–6 months. **Modulation = the volume knob** (turns pain up or down in the cord).

> ⚠ **Common confusion:** **Modulation** is the step where the signal is intensified or de-intensified *in the spinal cord* — not perception (which is the conscious "feeling" in the brain). Acute pain *correlates* with injury severity; chronic pain does *not*.

Practice Questions

1. Put the pain pathway in order.

   Show answer

   > Nociception → Transduction → Transmission → Modulation → Perception.

2. Which statement about acute pain is true?

   Show answer

   > Acute pain is protective — it warns of tissue damage and correlates with the severity of injury.

3. A diabetic patient describes burning, tingling pain in both feet. What type of pain is this?

   Show answer

   > Neuropathic pain (diabetic neuropathy — injury to peripheral nerves).

4. Which pain-pathway step involves intensifying or dampening the signal in the spinal cord?

Show answer

Modulation.

Clinical Scenario

Two patients have identical ankle fractures, but one rates pain 3/10 and the other 9/10. The new nurse is skeptical of the higher rating. How do you explain the difference?

Show answer

Pain is **subjective** — differences in substance P/glutamate levels, modulation in the spinal cord, prior experience, and emotional state mean the same injury produces different perceived pain. Pain is what the patient reports; both ratings are valid and should be treated.

☑ **Self-check:** Can you list the 5 pathway steps in order, contrast acute vs chronic and nociceptive vs neuropathic pain, and explain pain's subjectivity? If yes, move on.

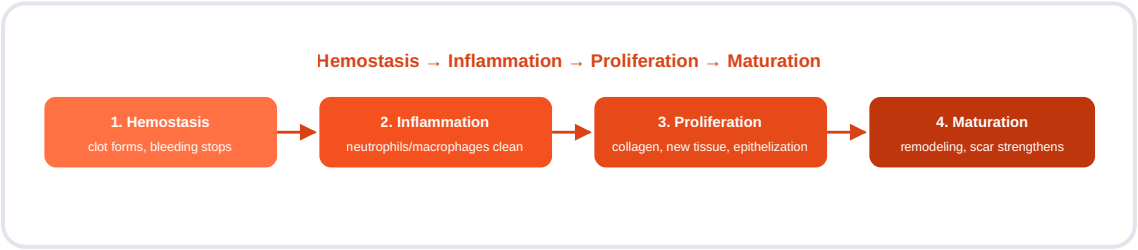
TOPIC 8

□Wound Healing: Phases & Outcomes

MUST KNOW ▼

Wound healing is the body's repair program. Know the **4 phases in order**, what happens in each, and the possible outcomes.

The 4 Phases of Healing



- **1. Hemostasis** — immediately: vasoconstriction + a clot (platelets, fibrin) stops bleeding and seals the wound.
- **2. Inflammation** — neutrophils then macrophages arrive to clean debris and bacteria; the classic redness/heat/swelling appears.
- **3. Proliferation** — **fibroblasts produce collagen**, new blood vessels form (angiogenesis), granulation tissue fills the gap, and **epithelialization** covers the surface.
- **4. Maturation (remodeling)** — collagen reorganizes and the scar gains strength over weeks to months.

Outcomes & Abnormal Scars

- **Regeneration** — tissue returns to normal (best outcome, possible in tissues that can regenerate).
- **Scar/fibrosis** — replaced by fibrous connective tissue (functional but not identical).
- **Keloid** — **excessive, raised scar tissue that grows beyond the original wound** borders (too much collagen).
- **Dehiscence** — wound edges separate/reopen; **evisceration** = organs protrude (surgical emergency).

Factors That Impair Healing

Poor circulation/oxygenation, diabetes (high glucose), infection, malnutrition (low protein/vitamin C/zinc), advanced age, smoking, and certain medications (steroids, chemotherapy).

Mnemonic: Phases = "**Healthy Is Pretty Much**" → **H**emostasis, **I**nflammation, **P**roliferation, **M**aturation. **Proliferation = "Producing"** (collagen + epithelialization). **Keloid = "Keeps growing"** beyond the wound.

⚠ **Common confusion: Epithelialization and collagen production = proliferation phase** (phase 3), not inflammation. A **keloid** (raised, excess scar beyond wound borders) is different from a normal/hypertrophic scar that stays within the wound.

Practice Questions

1. List the 4 phases of wound healing in order.

   Show answer

   Hemostasis → Inflammation → Proliferation → Maturation (remodeling).

2. A wound shows epithelialization and fibroblasts producing collagen. Which phase is this?

   Show answer

   Proliferation phase.

3. A month-old wound has a raised scar with excess tissue extending past the original borders. What is this?

   Show answer

   Keloid scarring.

4. **SATA:** Which factors impair wound healing? (A) Diabetes (B) Good protein intake (C) Smoking (D) Infection (E) Poor circulation

   Show answer

   A, C, D, E. Good protein intake *helps* healing.

Clinical Scenario

A diabetic patient's surgical incision is healing very slowly and becomes infected. Connect their diabetes to the impaired healing.

Show answer

High blood glucose impairs circulation and immune cell (neutrophil) function, and reduces oxygen/nutrient delivery to the wound. This slows the proliferation phase and raises infection risk, stalling healing.

☑ **Self-check:** Can you order the 4 healing phases, identify the proliferation phase by its features, define keloid, and list factors that impair healing? If yes, move on.

TOPIC 9

Neurodegenerative I: Parkinson's, Alzheimer's & Huntington's

MUST KNOW

These diseases destroy nervous tissue over time. The key to the exam is matching each disease to its **pathophysiology** (what's breaking down) and its **hallmark signs/symptoms**.

Parkinson's Disease

Patho: Loss of **dopamine**-producing cells in the **substantia nigra** → movement problems.
Signs (movement): **B**radykinesia (slow movement), **resting tremor**, **rigid muscles**, **shuffling gait**, stooped posture. Cognition is usually preserved until late.

Alzheimer's Disease

Patho: **Amyloid plaque** (and tau tangle) formation in the brain → progressive memory loss and personality change. Progressive and incurable.
Signs — the 5 A's: **A**mnesia (memory loss), **A**phasia (loss of speech/language), **A**praxia (loss of coordinated/purposeful movement), **A**gnosia (can't interpret sensations/recognize things), **A**nomia (can't name/retrieve words).

Huntington's Disease

Patho: Inherited, **autosomal dominant** disorder. No cure.
Signs: **Chorea** (involuntary, jerky movements of arms, legs, face), progressive loss of cognition, memory loss, dementia, emotional lability. Symptoms usually begin **mid-life (30s–40s)**.

Mnemonic: **Parkinson's = "no Dopamine, so you can't move smoothly"** (think P = "Poverty of movement," substantia **n**igra). **Alzheimer's = Amyloid + the 5 A's. Huntington's = "Hunting = chasing/jerky" chorea, and it's Handed down (autosomal Dominant).**

⚠ **Common confusion: Parkinson's = resting tremor + slow/rigid movement** (dopamine loss). **Huntington's = involuntary excess movement (chorea)** and is inherited (autosomal dominant). **Alzheimer's = amyloid plaques + memory/ the 5 A's.** Match patho to disease — that's how the exam tests it.

Practice Questions

1. What is the pathophysiology of Parkinson's disease?

   Show answer

   > Loss of dopamine-producing cells in the substantia nigra, causing movement-related symptoms.

2. Name the 5 A's of Alzheimer's disease.

   Show answer

   > Amnesia, Aphasia, Apraxia, Agnosia, Anomia.

3. A client has involuntary jerky movements and a family history of the same disease starting in their 30s. Likely diagnosis and inheritance pattern?

   Show answer

   > Huntington's disease — autosomal dominant; chorea is the hallmark.

4. **SATA:** Which would you expect in Alzheimer's? (A) Difficulty coordinating movement (apraxia) (B) Loss of ability to express/understand speech (aphasia) (C) Inability to retrieve words (anomia) (D) Resting tremor as the first sign (E) Inability to interpret sensations (agnosia)

   Show answer

   > **A, B, C, E.** Resting tremor as a first sign points to Parkinson's, not Alzheimer's.

Clinical Scenario

A newly diagnosed Parkinson's patient asks the nurse for safety advice at home. Given the pathophysiology and symptoms, what nursing considerations matter most?

Show answer

Because of bradykinesia, rigidity, shuffling gait, and stooped posture, **fall prevention** is key: remove rugs/clutter, install grab bars, allow extra time for movement, use assistive devices, and watch for swallowing/aspiration risk as it progresses.

☑ **Self-check:** Can you give the patho + hallmark signs for Parkinson's (dopamine/tremor), Alzheimer's (amyloid/5 A's), and Huntington's (autosomal dominant/chorea)? If yes, move on.

TOPIC 10

⚡ Neurodegenerative II: MS, ALS & Guillain-Barré

MUST KNOW

These three are easy to mix up. The trick: **what is being destroyed, and where (CNS vs PNS)?**

Multiple Sclerosis (MS)

Patho: Progressive destruction of the **myelin sheath in the CNS** (brain & spinal cord). Cause unknown (autoimmune-linked). Relapsing-remitting course.
Signs: fatigue, **vision loss/blurred vision**, **incontinence**, balance & gait problems, loss of cognition.

Amyotrophic Lateral Sclerosis (ALS)

Patho: Progressive loss of **motor neurons** → loss of skeletal muscle function.
Signs: skeletal muscle wasting, progressive paralysis, declining respiratory & swallowing function. **Crucial point: NO brain impact — cognition and sensation remain intact.**

Guillain-Barré Syndrome (GBS)

Patho: **Ascending** polyneuropathy from **demyelination of the PNS**; typically **preceded by a viral infection**. Often reversible.
Signs: tingling and weakness that **start in the feet/legs and progress upward (centrally)**; can advance to loss of respiratory function, slurred speech, and paralysis.

Mnemonic: MS = Myelin Sheath, CNS ("MS = My Sheath, in my brain/cord"). **ALS = "All muscles Lose Strength" but brain is fine** (think Lou Gehrig — mind sharp, body fails). **GBS = "Goes Bottom-up, after a Sickness"** (ascending, PNS demyelination, post-viral).

⚠ **Common confusion: MS = demyelination in the CNS; GBS = demyelination in the PNS** (and GBS ascends, post-viral). **ALS spares**

cognition/sensation — that's the giveaway. If a question mentions ascending weakness after a recent infection, think GBS, not MS.

Practice Questions

1. A client has blurred vision, urinary incontinence, and gait issues. What is the pathophysiology behind MS?

   Show answer

   > Destruction of the myelin sheath in the central nervous system.

2. A client had a viral infection a week ago and now has paralysis ascending from the feet toward the waist. Likely diagnosis?

   Show answer

   > Guillain-Barré Syndrome (ascending, PNS demyelination, post-viral).

3. What distinguishes ALS from the other neurodegenerative diseases regarding the brain?

   Show answer

   > ALS spares the brain — cognition and sensory processing remain intact while motor function is lost.

4. **SATA:** Which signs would you expect in advanced ALS? (A) Progressive paralysis (B) Decreased respiratory function (C) Loss of cognition (D) Skeletal muscle wasting (E) Swallowing dysfunction

   Show answer

   > **A, B, D, E.** Loss of cognition is NOT a feature of ALS (cognition stays intact).

Clinical Scenario

Two patients have weakness. Patient A had a viral illness last week and now has weakness rising from the feet up. Patient B has had relapsing episodes of blurred vision and incontinence over several years. Which has GBS and which has MS, and why?

Show answer

Patient A = **Guillain-Barré** (acute, ascending, post-viral, PNS demyelination). Patient B = **MS** (chronic relapsing-remitting, CNS demyelination causing vision/ bladder/gait issues).

☑ **Self-check:** Can you state where demyelination occurs in MS vs GBS, the "ascending + post-viral" GBS pattern, and that ALS spares cognition? If yes, move on.

TOPIC 11 □Inflammatory & Autoimmune Diseases MUST KNOW ▼

Most of these are **autoimmune** — the immune system attacks the body's own tissue. Learn the patho, the hallmark sign, and the key lab/risk factor for each.

Gout

Patho: Buildup of **purines** → **uric acid** → **crystal formation** in joints → local inflammation and pain. Lab: **elevated uric acid**.
Hallmark: sudden, severely painful, red, swollen joint — classically the **big toe**.
Risk factors: alcohol, **red meat / purine-rich foods**, obesity, dehydration, family history.

Rheumatoid Arthritis (RA)

Patho: **Autoimmune** — antibodies attack the joints (synovium), typically the **small joints of hands, feet, wrists, bilaterally**. Over time inflammation destroys the joints.
Hallmark: **morning stiffness that improves as the day goes on**; symmetric/bilateral joint involvement.

Systemic Lupus Erythematosus (SLE)

Patho: Autoimmune — autoantibodies attack **nucleic acids (DNA/RNA)**; can affect almost every body system (multi-system).
Hallmark: **butterfly (malar) rash** across cheeks and nose; plus muscle/joint pain, fever, fatigue, possible kidney damage and anemia.

Scleroderma & CREST

Patho: Autoimmune disorder causing **endothelial damage** → **excess collagen and fibrosis**, hardening skin and organs.
CREST (limited scleroderma): **C**alcinosis, **R**aynaud's, **E**sophageal dysfunction, **S**clerodactyly, **T**elangiectasis.

Fibromyalgia

Patho: Chronic condition causing **widespread pain** throughout the body (amplified pain processing).
Signs: chronic pain, fatigue, **poor sleep**, impaired cognition ("fibro fog"), irritable bowel. Risk: stress, female sex, family history.

Mnemonic: Gout = "Great tOe + Uric acid." RA = morning stiffness that Relaxes As the day goes, bilateral small joints. **Lupus = butterfly + attacks the "nucleus" (DNA). Scleroderma = "Sclero = hard" skin from collagen; CREST** spells its 5 features. **Fibromyalgia = "Fibro = Fatigue + widespread pain + Fog."**

⚠ **Common confusion: RA stiffness improves through the day** (osteoarthritis tends to worsen with use). **Gout = uric acid crystals** (not autoantibodies). **Lupus targets nucleic acids (DNA/RNA)** — that's the classic "what's attacked" answer. Scleroderma's problem is **excess collagen/fibrosis from endothelial damage**.

Practice Questions

1. What lab value is elevated in gout, and what's the biggest dietary risk factor?

   Show answer

   > Uric acid is elevated; consuming red meats (purine-rich foods) is a major risk factor.

2. A client has morning wrist stiffness that improves during the day. Which condition?

   Show answer

   > Rheumatoid arthritis.

3. In SLE, what do the autoantibodies typically target?

   Show answer

   > Nucleic acids (DNA/RNA).

4. What does CREST stand for in limited scleroderma?

Show answer

Calcinosis, Raynaud's, Esophageal dysfunction, Sclerodactyly, Telangiectasis.

5. A client reports widespread chronic pain, fatigue, poor sleep, and "brain fog." Likely diagnosis?

   Show answer

   Fibromyalgia.

Clinical Scenario

A young woman has a facial rash across her cheeks and nose, joint pain, fatigue, and lab work showing anemia and early kidney damage. What condition fits, and why does it affect so many systems?

Show answer

Systemic Lupus Erythematosus (SLE). Because autoantibodies attack **nucleic acids (DNA/RNA)** found in every cell, lupus can damage nearly any organ — skin (butterfly rash), joints, blood (anemia), and kidneys.

☑ **Self-check:** Can you give the hallmark for gout (uric acid/big toe), RA (morning stiffness that improves, bilateral), SLE (butterfly rash/anti-DNA), scleroderma (collagen/CREST), and fibromyalgia (widespread pain/poor sleep)? If yes, move on.

TOPIC 12 □Neuromuscular Junction Diseases MUST KNOW ▼

The **neuromuscular junction (NMJ)** is where a nerve meets a muscle and releases **acetylcholine (ACh)** to trigger contraction. These three diseases all disrupt the NMJ — but in different ways and different **directions of paralysis**.

Botulism

Patho: *Clostridium botulinum* toxin **blocks the release of acetylcholine** at the NMJ → muscles can't fire.
Causes: improperly canned/preserved foods; honey (in infants).
Signs: **DESCENDING paralysis** (starts at the head/face and moves down), difficulty swallowing, airway paralysis, blurred/double vision.

Tetanus

Patho: *Clostridium tetani* toxin enters through an **open wound** and **blocks inhibitory nerves** → uncontrolled muscle **over-excitation**.
Signs: severe muscle **spasms**, **lockjaw** (trismus), seizures, can cause broken bones from the violent spasms.

Myasthenia Gravis (MG)

Patho: **Autoimmune** — loss/destruction of **ACh receptors** at the NMJ. Chronic, relapse-and-remission.
Signs: ocular first — **ptosis** (drooping eyelids) and **diplopia** (double vision); progressive muscle weakness affecting face, lips, tongue, neck, throat; **worse with use / end of day**; risk of respiratory muscle weakness.

□**Mnemonic:** **Botulism = "Blocks ACh = limp" + Descending** ("B comes down from the top"; canned food & honey). **Tetanus = "Tightens" (over-excitation), lockjaw, from a wound** (rusty nail). **Myasthenia Gravis = "Muscles Grow tired" — Ptosis + worse at end of day**; autoimmune loss of ACh receptors.

⚠ **Common confusion: Botulism = DEScending paralysis & blocks ACh release** (limp/floppy). **Guillain-Barré = AScending** (don't mix them up!). **Tetanus = rigid/spasms** (over-excitation), opposite of botulism's limpness. **MG = receptor problem, worsens with activity.**

Practice Questions

1. What is the pathophysiology of botulism?

   Show answer

   > The toxin blocks the release of acetylcholine at the neuromuscular junction, causing descending paralysis.

2. An unvaccinated client steps on a rusty nail and a week later has muscle spasms and lockjaw. Diagnosis?

   Show answer

   > Tetanus (toxin blocks inhibitory nerves → over-excitation/spasms).

3. A client has eyelid drooping and double vision that worsen by the end of the day. Likely diagnosis and patho?

   Show answer

   > Myasthenia Gravis — autoimmune loss of ACh receptors at the NMJ; weakness worsens with use.

4. Which question best helps confirm botulism in a patient with descending paralysis and double vision?

   Show answer

   > "Do you can or preserve your own foods?" (improper canning is a classic botulism source).

Clinical Scenario

Two patients both have weakness and breathing trouble. Patient A ate home-canned green beans and has paralysis starting in the face moving downward. Patient B has had years of fluctuating eyelid droop and double vision worse at night. Which is botulism and which is myasthenia gravis?

Show answer

Patient A = **botulism** (acute, food-borne toxin blocking ACh release, descending paralysis). Patient B = **myasthenia gravis** (chronic autoimmune loss of ACh receptors; ptosis/diplopia worse with use). Both threaten the respiratory muscles — monitor breathing closely.

☑ **Self-check:** Can you contrast botulism (blocks ACh, descending, limp) vs tetanus (over-excitation, rigid, lockjaw) vs MG (autoimmune receptor loss, worse with use), and distinguish botulism's descending vs GBS's ascending paralysis? If yes — all 12 topics done!

□Exam Cheat Sheet

□MODULE 3 — Cell Injury, Cancer & Inflammation

- **Hypoxia = #1 cause of cell injury.** Effects: ↓ATP, Na/K pump fails, cell swelling, no protein synthesis.
- **Ischemia** = short-term lack of blood flow, no cell death (reversible) → infarction if prolonged. **Hypoxemia** = low O&sub2; in blood.
- **Adaptations:** Hypertrophy (bigger), Atrophy (smaller), Hyperplasia (more), Metaplasia (swap type, reversible), **Dysplasia (disordered = pre-cancer)**.
- **Free radicals** = unstable, reactive, oxidative damage; ↑ with carcinogens; neutralized by antioxidants.
- **Cancer naming:** Carcinoma=epithelial, Sarcoma=connective, Glioma=brain, Lymphoma=lymph, Melanoma=pigment, Leukemia=WBC, Myeloma=plasma.
- **Malignant** = rapid, poorly differentiated, invades, metastasizes, angiogenesis. **Benign** = slow, well differentiated, local.
- **Grading = how cells look** (differentiation). **Staging = how far spread** (I local → IV distant; TNM).
- **Metastasis sites:** brain, bones, lungs, liver. **Screening:** PAP=cervix, mammogram=breast, colonoscopy=colon, PSA=prostate.
- **Oncogenic viruses:** HPV, Hep B/C, HSV, EBV.
- **Inflammation:** mast cells → histamine → vasodilation (redness) + permeability (swelling). **Neutrophils first.** Opsonization = tagging for engulfment.
- **SIRS → Sepsis (+infection) → Severe sepsis (+organ damage) → Septic shock (+low BP).**

□MODULE 4 — Pain & Wounds

- **Pain pathway:** Nociception → Transduction → Transmission → Modulation → Perception. (Modulation = amplify/dampen in spinal cord.)
- **Acute pain = protective**, correlates with injury. **Chronic** >3–6 mo, non-protective, no correlation.
- **Pain types:** Somatic (skin/muscle/bone), Visceral (organs, referred), Neuropathic (nerve injury — burning/tingling).
- **Pain is subjective** — substance P/glutamate differences, modulation, experience.
- **Wound healing:** Hemostasis → Inflammation → Proliferation (collagen + epithelialization) → Maturation.
- **Keloid** = raised scar beyond wound borders. Impaired by diabetes, smoking, infection, poor circulation, malnutrition.

□MODULE 4 — Neurodegenerative Diseases

- **Parkinson's:** ↓dopamine in substantia nigra → bradykinesia, resting tremor, rigidity, shuffling gait.
- **Alzheimer's:** amyloid plaques → memory loss. **5 A's:** Amnesia, Aphasia, Apraxia, Agnosia, Anomia.
- **Huntington's:** autosomal dominant → chorea (involuntary movements), dementia; onset 30s–40s.
- **MS:** myelin destruction in **CNS** → vision loss, incontinence, gait issues; relapsing.
- **ALS:** motor neuron loss → muscle wasting, paralysis; **cognition/sensation INTACT.**
- **Guillain-Barré: ascending** demyelination of **PNS**, post-viral.

□MODULE 4 — Autoimmune & NMJ Diseases

- **Gout:** purines → ↑uric acid → crystals; big toe; risk = red meat, alcohol, dehydration.
- **RA:** autoimmune, bilateral small joints, morning stiffness that **improves** during the day.
- **SLE:** autoantibodies vs **nucleic acids (DNA/RNA)**; butterfly rash; multi-system.
- **Scleroderma:** endothelial damage → excess collagen/fibrosis. **CREST** = Calcinosis, Raynaud's, Esophageal, Sclerodactyly, Telangiectasis.
- **Fibromyalgia:** widespread chronic pain, fatigue, poor sleep, fog.
- **Botulism:** blocks ACh release → **DEScending** paralysis; canned food/honey.
- **Tetanus:** blocks inhibitory nerves → over-excitation, spasms, lockjaw; via wound.
- **Myasthenia Gravis:** autoimmune loss of **ACh receptors**; ptosis/diplopia, worse with use.

□Quick Disease Comparison

"Don't Mix These Up" — high-yield distinctions

Pair	Key difference
Botulism vs Guillain-Barré	Botulism = **DEScending** & limp; GBS = **AScending**, post-viral.
Botulism vs Tetanus	Botulism = floppy (blocks ACh); Tetanus = rigid spasms (blocks inhibition).
MS vs ALS	MS = CNS myelin, vision/bladder; ALS = motor neurons, cognition intact.
Parkinson's vs Huntington's	Parkinson's = too little movement (tremor/rigid); Huntington's = too much (chorea).
Grading vs Staging	Grade = how cells look; Stage = how far spread.
Hypertrophy vs Hyperplasia	Trophy = size (bigger); Plasia = number (more).
Metaplasia vs Dysplasia	Metaplasia = reversible cell-type swap; Dysplasia = disordered, pre-cancer.

□Glossary

Cell, Cancer & Inflammation Terms

- **Hypoxia** — low oxygen in tissues; #1 cause of cell injury.
- **Ischemia** — short-term lack of blood flow, no cell death yet.
- **Infarction** — tissue death from prolonged ischemia.
- **Hypertrophy / Hyperplasia** — bigger cells / more cells.
- **Metaplasia** — reversible swap of one cell type for another.
- **Dysplasia** — disordered, pre-cancerous growth.
- **Free radical** — unstable, reactive molecule causing oxidative damage.
- **Carcinogen** — substance that damages DNA to promote cancer.
- **Metastasis** — spread of cancer to distant sites.
- **Grading / Staging** — cell appearance / extent of spread.
- **Opsonization** — tagging antigens so phagocytes can engulf them.
- **SIRS** — systemic inflammatory response criteria (vital signs).
- **Sepsis** — SIRS plus a confirmed infection.

Pain & Wound Terms

- **Nociception** — detection of a painful stimulus (step 1 of pathway).
- **Modulation** — amplifying or dampening the pain signal in the spinal cord.
- **Perception** — conscious awareness of pain in the brain.
- **Somatic / Visceral / Neuropathic** — skin-muscle-bone / organ / nerve pain.
- **Proliferation phase** — wound stage with collagen production & epithelialization.
- **Keloid** — raised scar extending beyond the original wound.
- **Dehiscence / Evisceration** — wound reopening / organs protruding.

Neurodegenerative Terms

- **Substantia nigra** — dopamine-producing brain region lost in Parkinson's.
- **Bradykinesia** — abnormally slow movement (Parkinson's).
- **Chorea** — involuntary jerky movements (Huntington's).
- **Amyloid plaque** — abnormal protein deposits in Alzheimer's.
- **The 5 A's** — Amnesia, Aphasia, Apraxia, Agnosia, Anomia.
- **Demyelination** — loss of myelin sheath (MS = CNS, GBS = PNS).
- **Ascending paralysis** — weakness rising from feet upward (GBS).

Autoimmune & NMJ Terms

- **Autoimmune** — immune system attacks the body's own tissue.
- **Uric acid** — elevated in gout; forms joint crystals.
- **Malar (butterfly) rash** — classic facial rash of SLE.
- **CREST** — Calcinosis, Raynaud's, Esophageal dysfunction, Sclerodactyly, Telangiectasis.
- **Acetylcholine (ACh)** — neurotransmitter at the neuromuscular junction.
- **Ptosis / Diplopia** — drooping eyelid / double vision (myasthenia gravis).
- **Descending paralysis** — weakness from head downward (botulism).

Final Study Tips

- **Master the "patho → disease" match.** Most exam questions describe a patho or symptom set and ask for the disease. Drill the hallmark of each one.
- **Memorize the ordered lists:** pain pathway (5 steps), wound healing (4 phases), the 5 A's, and the SIRS→sepsis→shock ladder.
- **Lock in the "don't mix these up" table** — botulism vs GBS (descending vs ascending), MS vs ALS, grading vs staging, hypertrophy vs hyperplasia.
- **Practice SATA questions carefully.** Read each option as its own true/false statement — partial credit usually isn't given.
- **Use the cheat sheet** the night before AND the morning of the exam, then do a final self-test with the answers hidden.
- **Remember the format:** 50 questions, proctored with Respondus LockDown Browser — set up your space and whiteboard ahead of time.

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You've got this. Learn the cell's story first, and every disease becomes a chapter you already understand.

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